

SENTIMENT DATASET

In Week 6, we explored Logistic Regression as a supervised machine learning algorithm for binary classification. The primary objective was to apply Logistic Regression to a textual and sentiment dataset in order to classify the directional stock price. Given the broader goal of predicting stock price direction (up or down) using sentiment signals, Logistic Regression serves as an interpretable and computationally efficient baseline model.

The Sentiment Dataset contains the following columns:

- `full_text` – Combined news title and content
- `sentiment_neg` – Negative sentiment score
- `sentiment_pos` – Positive sentiment score
- `sentiment_neu` – Neutral sentiment score

The target variable for modeling is stock price direction (Up/Down), derived from daily closing price movement. The dataset combines structured numerical sentiment scores with unstructured textual data, allowing for both feature-engineered and text-vectorized modeling approaches. The final feature matrix consisted of:

- TF-IDF vectors from `full_text`
- Three sentiment score columns

Logistic Regression was trained using L2 regularization to prevent overfitting, particularly important given the high dimensionality of TF-IDF features.

Key modeling considerations included:

- **Regularization strength (C parameter):** Controls the tradeoff between bias and variance.
- **Class balance:** Checked to ensure no strong imbalance between up and down days.
- **Feature scaling:** Not required for TF-IDF features but considered for sentiment scores if necessary

The hyperparameter-tuned Logistic Regression model demonstrated superior performance compared to the baseline model under identical data conditions. Improvements were observed in accuracy, precision, recall, and F1-score, confirming that tuning regularization strength is critical in high-dimensional TF-IDF settings. Nevertheless, the gains were incremental rather than dramatic, reinforcing that while hyperparameter tuning optimizes model performance, more complex nonlinear models may be required to capture deeper patterns within sentiment-driven financial prediction tasks.

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Logistic Regression (Combined Sentiment Dataset)

C	Max Iterations	Training Accuracy	Training F1	Validation Accuracy	Validation F1
10	200	76.417%	0.748	56.175%	0.557
10	200	76.500%	0.749	57.769%	0.574
10	500	76.417%	0.748	56.175%	0.557
10	500	76.500%	0.749	57.769%	0.574
10	1000	76.417%	0.748	56.175%	0.557
10	1000	76.500%	0.749	57.769%	0.574
1	200	70.583%	0.680	53.386%	0.498
1	200	70.583%	0.680	53.386%	0.498
1	500	70.583%	0.680	53.386%	0.498
1	500	70.583%	0.680	53.386%	0.498
1	1000	70.583%	0.680	53.386%	0.498
1	1000	70.583%	0.680	53.386%	0.498
0.1	200	54.232%	0.405	52.988%	0.367
0.1	200	54.314%	0.405	52.988%	0.367
0.1	500	54.232%	0.405	52.988%	0.367
0.1	500	54.314%	0.405	52.988%	0.367
0.1	1000	54.232%	0.405	52.988%	0.367
0.1	1000	54.314%	0.405	52.988%	0.367
0.01	200	52.835%	0.365	52.988%	0.367
0.01	200	52.835%	0.365	52.988%	0.367
0.01	500	52.835%	0.365	52.988%	0.367
0.01	500	52.835%	0.365	52.988%	0.367
0.01	1000	52.835%	0.365	52.988%	0.367
0.01	1000	52.835%	0.365	52.988%	0.367

HISTORICAL PRICE DATASET

For the historical price dataset, we used a logistic regression model as a baseline model. In this dataset, only lagged values of stock price direction will be used to predict whether AAPL stock will go up or down on a given day. This is a relatively small dataset with only a few predictor variables.

A logistic regression model was fitted using the training data from the historical price data using specified hyperparameters of 500 iterations and a regularization strength (C) of 1. The metrics of this initial model were noted and used to tune hyperparameters. For consistency across the combined and historical price datasets, we used a five-fold cross validation grid search to see how changing the number of iterations changed the model's performance. We will evaluate performance using accuracy and F1-score. C values tested were 0.1, 1, and 5, and iteration counts were 200, 500, and 1,000. Results for the 9 possible combinations were recorded as follows.

Logistic Regression (Historical Price Dataset)

C	Max Iterations	Training Accuracy	Training F1	Test Accuracy	Test F1
5	200	52.262%	0.635	50.613%	0.625
5	500	52.262%	0.635	50.613%	0.625
5	1000	52.262%	0.635	50.613%	0.625
1	200	52.262%	0.635	50.613%	0.625
1	500	52.262%	0.635	50.613%	0.625
1	1000	52.262%	0.635	50.613%	0.625
0.1	200	53.753%	0.648	50.695%	0.636
0.1	500	53.753%	0.648	50.695%	0.636
0.1	1000	53.753%	0.648	50.695%	0.636

In this case, tuning hyperparameters did very little to change the model's performance. While lowering C improves accuracy and F1-score slightly, it is not a significant difference. There is also a noticeable difference between training and validation accuracy. The strongest performing combination is with C=0.1 and 200 iterations. In this case the validation accuracy is just over 50% a 50.695% and the F1-score is 0.626. Overall, this is not a strong model for predicting stock price direction. This may be a result of the dataset containing too few predictor variables. Going forward, we will need to examine how this dataset fares with other models and how well models built with the other datasets can predict stock price direction.

COMBINED DATASET

The model choice for this week is the logistic model, which will attempt to classify daily observations into whether AAPL stock price increased or decreased using price timeseries and news sentiment data. The logistic model will serve as a baseline with which we can compare future models with. It is a simpler model, and will be a good benchmark for the complex models we will fit in future weeks.

After an initial logistic model was fit on the training set, it was evaluated on the validation set. The initial model had a max iteration number of 500, and a ‘C’ regularization strength of 1. These parameters are both important since regularization can control how closely the model will fit the training data and the max iteration can help the model reach convergence. Both these hyper-parameters were varied in a 5-fold cross validation grid search, using values below and above the default settings. The primary metrics used to evaluate performance were accuracy and F1 score. Accuracy is important because this is the metric we initially used to estimate project value, and F1 score is good to balance any imbalances between the majority “up” and minority “down” classes.

Logistic Regression (Combined Dataset)

C	Max Iterations	Training Accuracy	Training F1	Test Accuracy	Test F1
5	200	55.115%	0.642	50.990%	0.602
5	500	55.115%	0.642	50.990%	0.602
5	1000	55.115%	0.642	50.990%	0.602
1	200	54.971%	0.644	50.164%	0.608
1	500	54.971%	0.644	50.164%	0.608
1	1000	54.971%	0.644	50.164%	0.608
0.1	200	54.187%	0.654	50.161%	0.623
0.1	500	54.187%	0.654	50.161%	0.623
0.1	1000	54.187%	0.654	50.161%	0.623

Across the nine possible combinations of hyper-parameters, max iterations had no effect on performance, but the C regularization parameter did. Training accuracy was the highest for regularization parameter of 5, and test accuracy was also highest when it equaled 5. Training F1 was highest when the regularization parameter equaled 0.1, and test F1 was also highest when it equaled 0.1. Overall, validation set performance was somewhat close to training, but it did underperform slightly, which indicates some overfitting.

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Choosing the best model is difficult because the best accuracy and F1 scores occurred on different regularization hyperparameter values. However, since accuracy was initially used to show monetary value of the model, this will be the most important metric. The best model for this week is the model with a C regularization value of 5 and a maximum iteration value of 200, resulting in a test accuracy of 50.99% and an F1 score of 0.602.